



Comparing dynamical systems with Optimal Transport

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SGM seminar

1. Estimating & Interpreting representations of dynamical systems
2. Comparing dynamical systems:
Spectral-Grassmann Wasserstein Metric (SGOT)
3. Experiments

Estimating and interpreting
representation of dynamical systems

Dynamical system & machine learning

Dynamical systems are backbone models of temporally evolving phenomena:

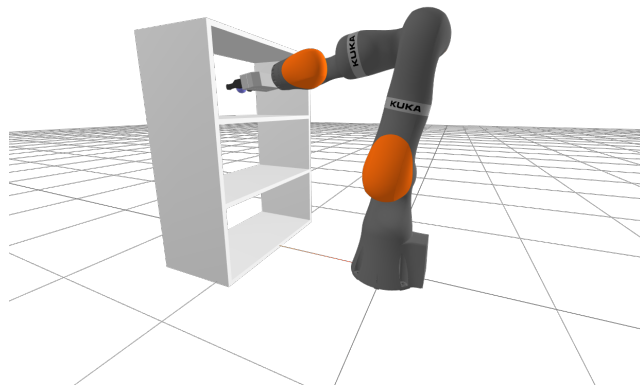
Modeling:

$$x_{t+1} = g(t, x_t)$$

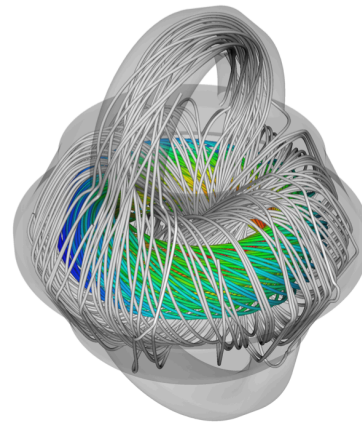
discrete time

$$\dot{x}(t) = g(t, x(t))$$

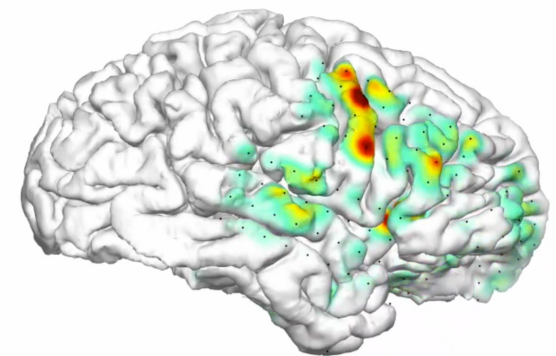
continuous time



Robotic



Fluid dynamic



Brain dynamic

Paradigm shift:

- Classical approach: ODE/PDE/SDE design + parameter fitting
- ML approach : Can we build dynamical models purely from observed data ?

➡ Elegant formulation through transfer operator theory.

Linear autonomous dynamical systems

Modeling:

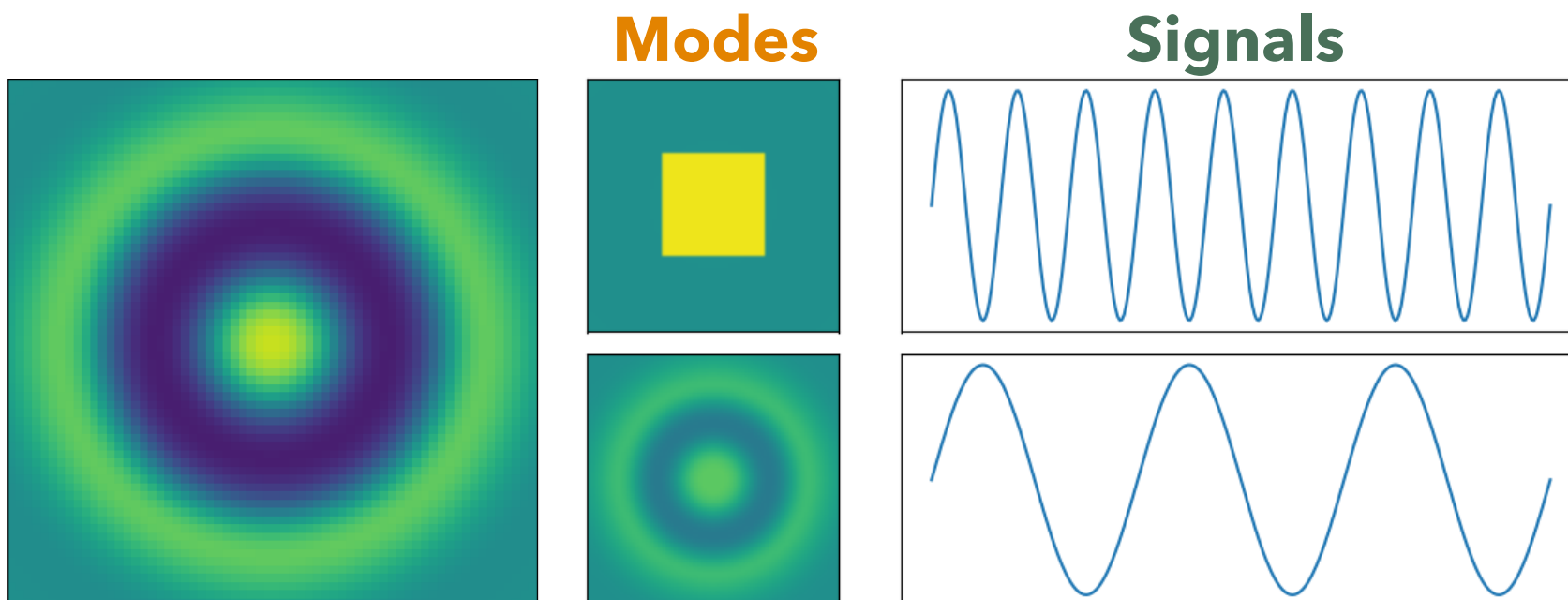
$$\dot{x}(t) = Ax(t) \rightarrow x(t) = \exp(tA)x_0$$

Non-defective system (spectral decomposition):

$$A = \underbrace{\begin{pmatrix} | & & | \\ p_1 & \cdots & p_n \\ | & & | \end{pmatrix}}_P \underbrace{\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}}_\Lambda \underbrace{\begin{pmatrix} - & p_1^{-1} & - \\ & \vdots & \\ - & p_n^{-1} & - \end{pmatrix}}_{P^{-1}}$$

(Superposition principle)
$$x(t) = \sum_{k=1}^n \boxed{e^{t\lambda_k}} \boxed{p_k \langle p_k^{-1}, x_0 \rangle} = \sum_{k=1}^n \boxed{e^{t\tau_k + i2\pi\omega_k t}} \boxed{m_k(x_0)}$$

➔ Sum of elementary solutions that decouple **temporal** and **static** components.



- $\tau_k = \mathcal{R}(\lambda_k)$ timescale.
- $\omega_k = \mathcal{I}(\lambda_k)/2\pi$ frequency.
- $m_k(x_0) = \langle p_k^{-1}, x_0 \rangle p_k$ **mode**.

Non-linear autonomous dynamical systems

Modeling:

$$\dot{x}(t) = g(x(t)) \quad \rightarrow \quad x(t) = S_t(x_0) = x_0 + \int_0^t g(x(\tau)) d\tau$$

Koopman operator: Suppose a function space $\mathcal{F} \subset \mathbb{R}^{\mathcal{X}}$ (observables), for any time $t > 0$, the Koopman operator $T_t : \mathcal{F} \rightarrow \mathcal{F}$ associated to the dynamic verifies:

$$[T_t f](x_0) \triangleq f(S_t(x_0)) \quad \forall f \in \mathcal{F}, \forall x_0 \in \mathcal{X}$$

Remarks: Koopman operators are linear:

$$T_t(f + cg) = T_t f + cT_t g$$

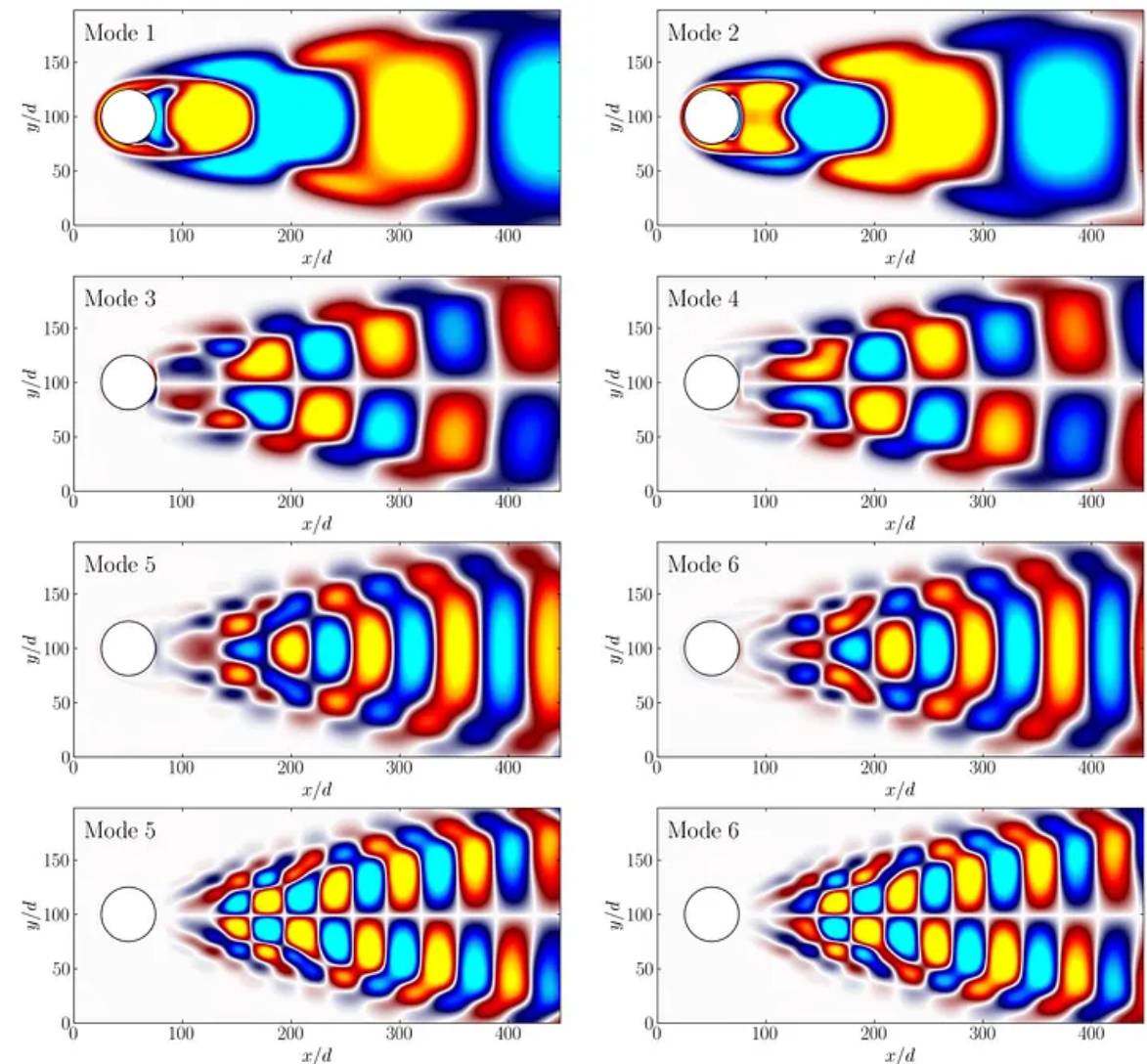
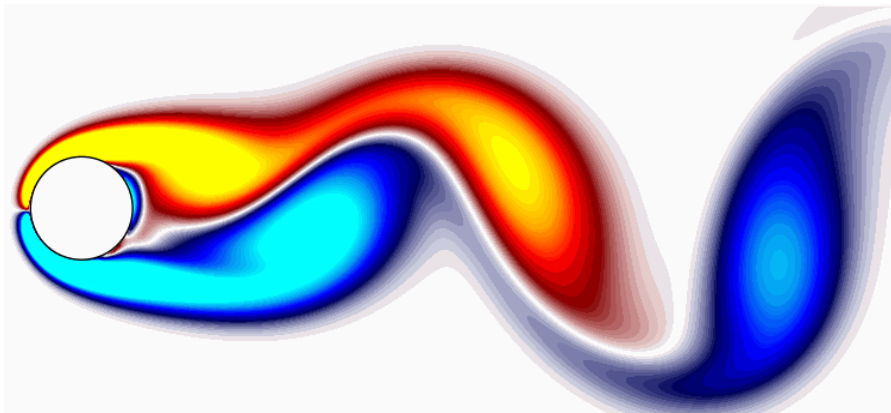
Infinitesimal generator: The semi-group of operators $\{T_t\}_{t>0}$ (i.e. $T_{s+t} = T_s T_t$) admits an infinitesimal generator such that:

$$A = \lim_{t \rightarrow 0} (T_t - Id)/t \quad \text{and} \quad T_t = \exp(tA)$$

Operator spectral decomposition

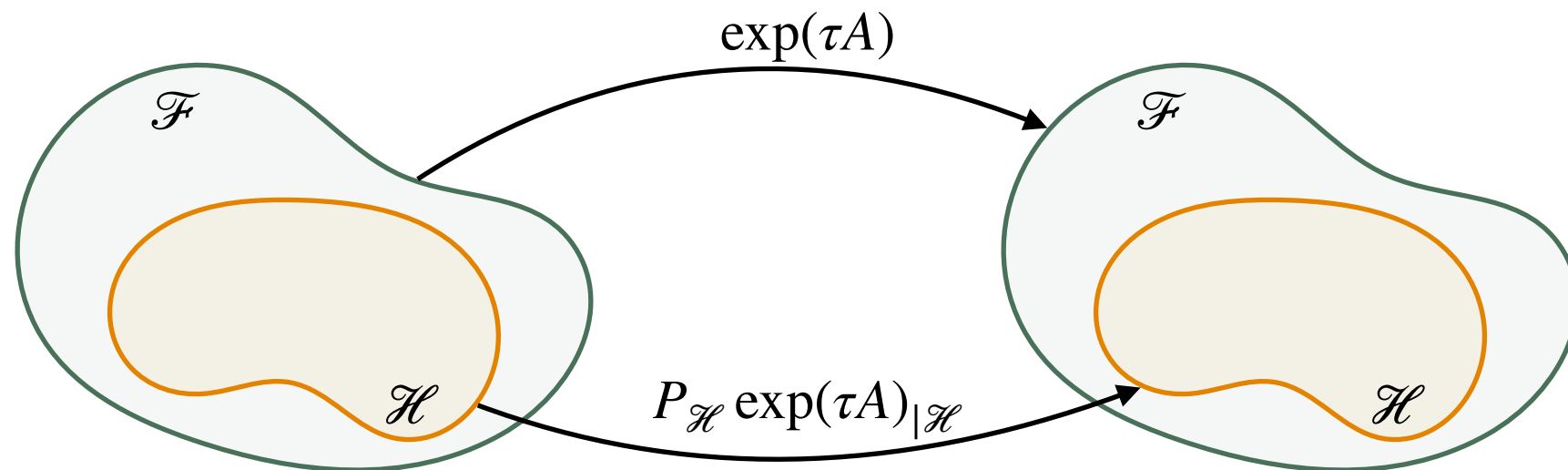
Spectral decomposition: Assuming that $\mathcal{F}$ is a separable Hilbert space (typically $L^2_\pi(\mathcal{X})$ with π the system's invariant measure) and A is a non defective operator with purely discrete spectrum. Then A can be written as:

$$Af = \sum_{j \in \mathbb{N}} \lambda_j \langle g_j, f \rangle f_j \quad \text{s.t.} \quad Lf_j = \lambda_j f_j, \quad L^*g_j = \bar{\lambda}_j g_j, \quad \langle f_i, g_j \rangle = \delta_{i,j}$$



Learning transfer operators

Problem: Only trajectories of dynamical systems are observed. Neither the operator A nor its domain $\mathcal{F}$ are known.



Popular approach [Bru22, Kos22]: Learning the action of $A : \mathcal{F} \rightarrow \mathcal{F}$ on a predefined RKHS $\mathcal{H} \subset \mathcal{F}$ with kernel $\kappa(x, y) \triangleq \langle \phi(x), \phi(y) \rangle$ via empirical risk minimization of the projected operator $P_{\mathcal{H}} \exp(\tau A)|_{\mathcal{H}}$, i.e.:

$$\operatorname{argmin}_{G \in HS(\mathcal{X})} \frac{1}{n} \sum_{i \in [n]} \|\phi(y_i) - G^* \phi(x_i)\|_{\mathcal{H}}^2$$

with the observation dataset $\{(x_i, y_i)\}_{i \in [n]}$ (typically a trajectory $y_i = x_{i+1}$).

Remark: Spectral estimation guarantees for universal bounded kernels [Kos23].

Transfer operators in practice

Classical approaches (low rank estimation):

- Dynamic Mode Decomposition (DMD) (linear kernel) [Kut16, Bru22].
- General kernel methods [Wil14, Kaw16].
- Reduced rank operator estimation [Kos22].
- Neural network approaches [Lus8, Kos24].

Open Problem:

How to compare transfer operators ?

Related work:

- Martin distance [Mar02] : pseudo-metric on ARMA models
- Binet-Cauchy kernel [Vid13]: Martin distance extension to LDS
- Topological conjugacy [Red24]: Optimal Transport on spectrum
- Hilbert-Schmidt and operator norms are too conservative

➡ Introducing Spectral-Grassmann Wasserstein Metric (SGOT)

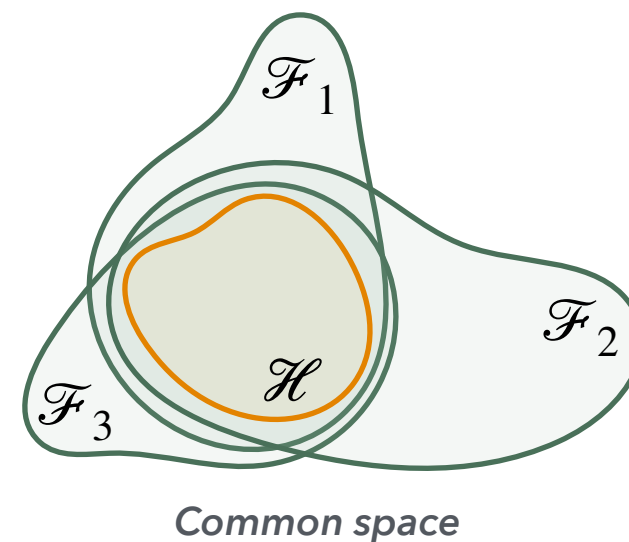
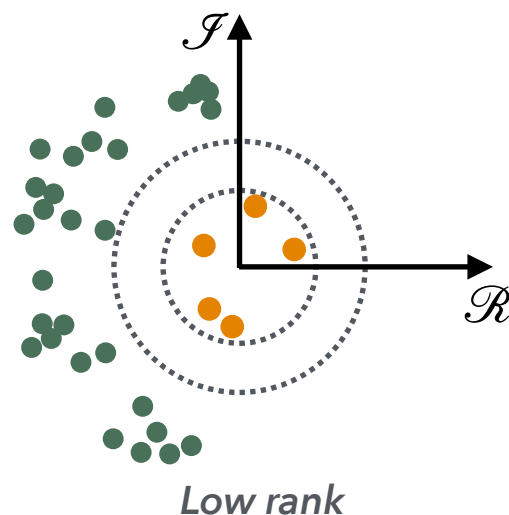
Comparing dynamical systems: Spectral-Grassmann Wasserstein Metric (SGOT)

SGOT intuition & assumptions

Intuition: Back to spectral decomposition:

$$Af = \sum_{i \in [\ell]} \sum_{j \in [m_i]} \lambda_i \langle g_{i,j}, f \rangle f_{i,j} = \sum_{i \in [\ell]} \lambda_i P_i f, \quad \text{s.t.} \quad \langle f_{i,j}, g_{u,v} \rangle = \delta_{(i,j),(u,v)}$$

- ➔ Comparing spectral components (sets of elementary solutions).
- ➔ **Invariances:** Decompositions are **unique up to a permutation** (optimal transport) and a **change of basis** (Grassmann manifold) for spectral projectors (P_i).



Assumptions:

1. **(Time homogeneous Markovian dynamical systems):** Let consider $\{A_k\}_{k \in [N]}$.
2. **(Low rank operators):** For any k , A_k has rank there exists $r_k < r_{max}$.
3. **(A common functional space):** There exists an RKHS $\mathcal{H}$ such that for any operator, the estimation of its r_k -restriction, $T_k = \exp(\tau A_k|_{r_k})$, is well defined.

Distributional embedding of operators

Intuition: Back to spectral decomposition:

$$Af = \sum_{i \in [\ell]} \sum_{j \in [m_i]} \lambda_i \langle g_{i,j}, f \rangle f_{i,j} = \sum_{i \in [\ell]} \lambda_i P_i f, \quad \text{s.t.} \quad \langle f_{i,j}, g_{u,v} \rangle = \delta_{(i,j),(u,v)} \text{ and } \sum_{i \in [\ell]} m_i = r$$

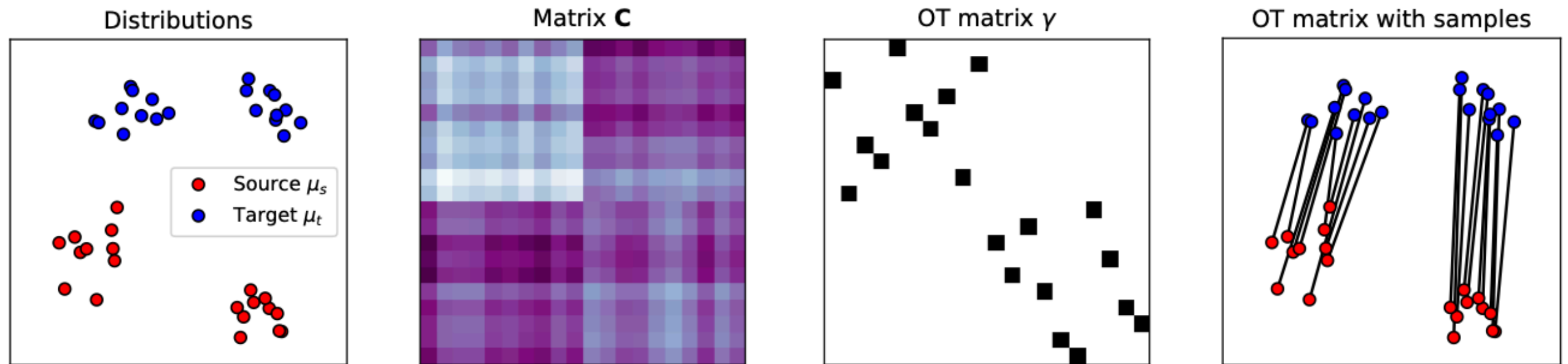
- m_i multiplicity of the eigenvalue λ_i .
- $P_i f \triangleq \sum_{j \in [m_i]} \langle g_{i,j}, f \rangle f_{i,j}$ the spectral projector of λ_i .
- $\mathcal{V}_i = \text{span}\{g_{i,j} \otimes f_{i,j}\}_{j \in [m_i]}$ the eigensubspace of λ_i .

Property (distributional embedding): The operator A can be represented as a probability distribution over the product space of eigenvalues and projectors:

$$\mu(L) \triangleq \sum_{i \in [\ell]} \frac{m_i}{r} \delta_{(\lambda_i, \mathcal{V}_i)}.$$

The embedding is injective and for bounded ranks such distributions can be compared with discrete Optimal Transport.

Discrete Optimal Transport



Kantorovich formulation:

Given two probability distributions $\mu_s = \sum_{i \in [n_s]} a_i \delta_{x_i^s}$ and $\mu_t = \sum_{i \in [n_t]} b_i \delta_{x_i^t}$, the optimal transport cost of order $p \in \mathbb{N}^*$ verifies:

$$W_{d,p}^p(\mu_s, \mu_t) = \min_{P \in \Pi(\mu_s, \mu_t)} \langle C, P \rangle_F,$$

where C is the cost matrix $c_{ij} = d(x_i^s, x_j^t)^p$, and the transport matrix P is from the set:

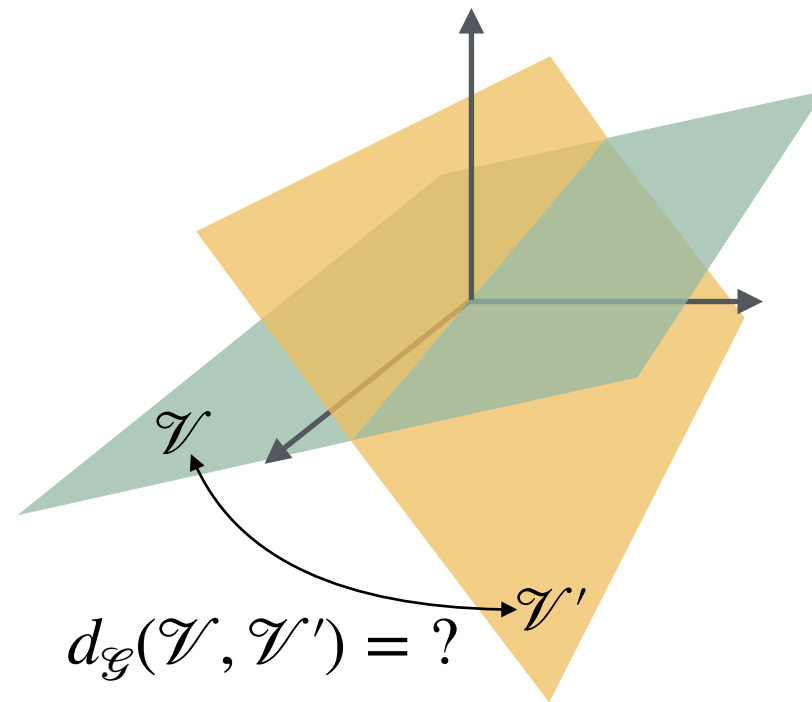
$$\Pi(\mu_s, \mu_t) = \{P \in \mathbb{R}_+^{n_s \times n_t} \mid P1_{n_t} = a, P^\top 1_{n_s} = b\}$$

Remarks:

- Computable with network simplex in $n^3 \log(n)$ where $n = n_s = n_t$.
- $W_d^p(\mu_s, \mu_t)$ is called a Wasserstein metric.

Grassmann Manifold

Aims: Comparing vector subspaces independently of their representative basis.



Proposition (Extension to infinite dimension space): Let $\mathcal{H}$ be a separable Hilbert space, and $\mathcal{G}_r(\mathcal{H})$ the set of vector subspaces of with dimension at most r . With the application:

$$d_G(\mathcal{V}, \mathcal{V}') = \|P_{\mathcal{V}} - P_{\mathcal{V}'}\|_{\mathcal{H}} \quad \text{for any } (\mathcal{V}, \mathcal{V}') \in \mathcal{G}(\mathcal{H})^2,$$

where P_* are orthogonal projectors, $\mathcal{G}_r(\mathcal{H})$ is a separable complete metric space.

Example: $d_G(\mathcal{V}, \mathcal{V}') = \sqrt{2(1 - \langle f, f' \rangle \langle g, g' \rangle)}$ with f, f', g and g' normalized.

Transfer operator as probability distribution

Spectral Grassmann Optimal Transport: Let $\mathcal{H}$ be a separable $\mathbb{C}$ -Hilbert space and $\mathcal{S}_r(\mathcal{H})$ the set of non-defective operator on $\mathcal{H}$ with rank at most $r \in \mathbb{N}$. For $P \geq 1$ and $\eta \in (0,1)$, the Spectral-Grassmann Wasserstein metric distance between two operators $A, A' \in \mathcal{S}_r(\mathcal{H})$:

$$d_{\mathcal{S}}(A, A') \triangleq W_{d_{\eta}, p}(\mu(A), \mu(A')),$$

where the cost function is defined by the metric:

$$d_{\eta}((\lambda, \mathcal{V}), (\lambda', \mathcal{V}')) \triangleq \eta |\lambda - \lambda'| + (1 - \eta) d_{\mathcal{G}}(\mathcal{V}, \mathcal{V}').$$

Then $(\mathcal{S}_r(\mathcal{H}), d_{\eta})$ is a metric space.

Computation:

- Pre-compute the cost matrix C with complexity $\mathcal{O}(r^2 n^2)$.
- Solve the OT problem with complexity $\mathcal{O}(r^3 \log(r))$ with network simplex.
- Overall complexity $\mathcal{O}(r^2 n^2 + r^3 \log(r))$.

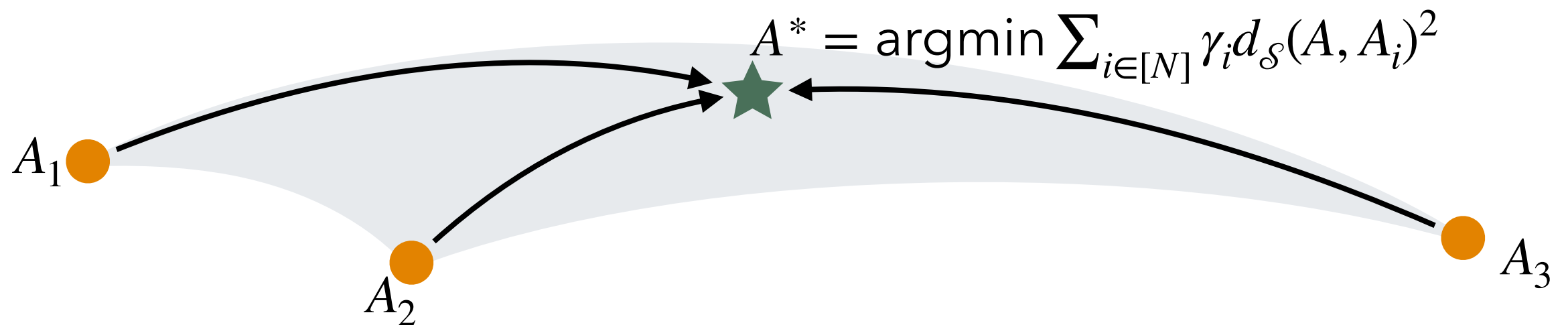
Metric estimator's converge bound

Recall: Under kernel universality assumption such methods come with spectral estimation guarantees [Kos23], specifically for the Reduced Rank Regression (RRR) method [Kos22].

Theorem (simplified): Suppose two dynamical systems with low rank operator projection $A, A' \in \mathcal{S}_r(\mathcal{H})$ and $\widehat{A}, \widehat{A}'$ their estimation from n samples with the RRR method [Kos22]. Suppose $\alpha \in (1,2)$, and $\beta \in [0,1]$ bounding the empirical covariance. In the i.i.d setting, for any $\delta \in (0,1)$, with probability $1 - \delta$ it holds:

$$|d_{\mathcal{S}}(\widehat{A}, \widehat{A}') - d_{\mathcal{S}}(A, A')| \leq n^{-\frac{\alpha-1}{2(\alpha+\beta)}} \ln(2\delta^{-1})$$

SGOT Barycenter



Fréchet mean problem: Given $\{A_i\}_{i \in [N]}$ and $\gamma \in \mathbb{R}_+^N$ s.t. $\sum_{i \in [N]} \gamma_i = 1$, we aim to solve:

$$\operatorname{argmin}_{A \in \mathcal{S}_r(\mathcal{H})} \sum_{i \in [N]} \gamma_i d_{\mathcal{S}}(A, A_i)^2$$

Constrained optimization problem: Considering the parametric operator:

$$A_{\theta} \triangleq \sum_{i \in [r]} \lambda_i \kappa_x \beta_i \otimes \kappa_x \alpha_i$$

with $\lambda \in \mathbb{C}^r$, $x \in \mathcal{X}^n$ and $\alpha, \beta \in \mathbb{C}^{n \times r}$, the problem becomes:

$$\operatorname{argmin}_{P, \theta} \sum_{i \in [N]} \gamma_i \langle C_i(\theta), P_i \rangle \quad \text{s.t.} \quad \begin{cases} K = \{\kappa(x_i, x_j)\}_{(i,j) \in [n]^2} \\ \alpha^* K \beta = I \\ P_i \in \Pi(\mu(A_{\theta}), \mu(A_i)), \forall i \in [N] \end{cases}$$

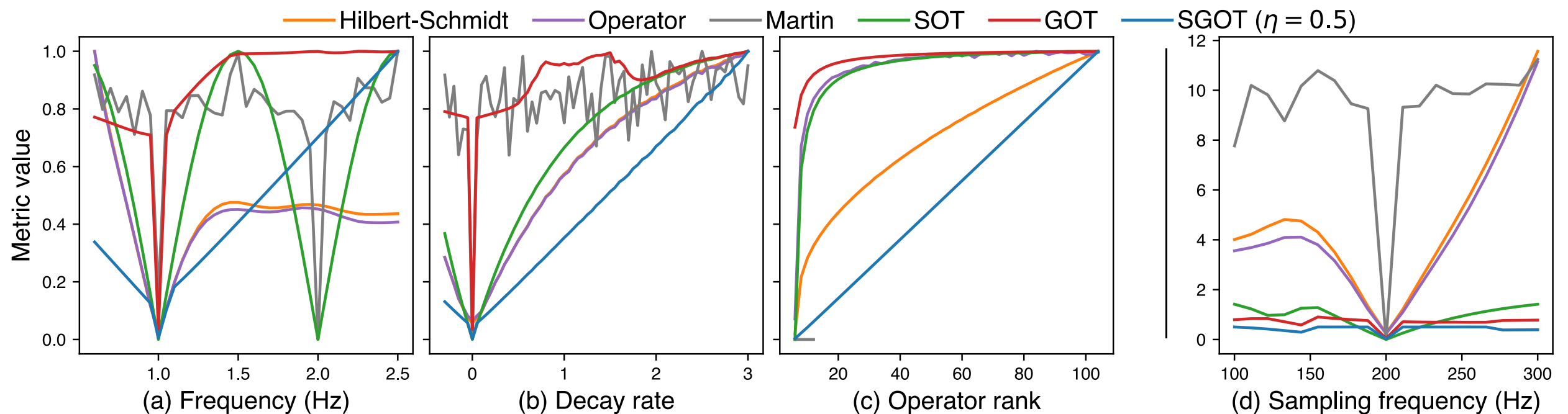
➔ *Optimization: Riemannian gradient descent or block coordinates descent.*

Experiments

Ablation study

Protocol: Metric comparisons between a baseline linear dynamical system and a shifted one under 4 scenarios:

- (a) Frequency shift.
- (b) Decay shift.
- (c) Rank shift.
- (d) Sampling frequency shift.



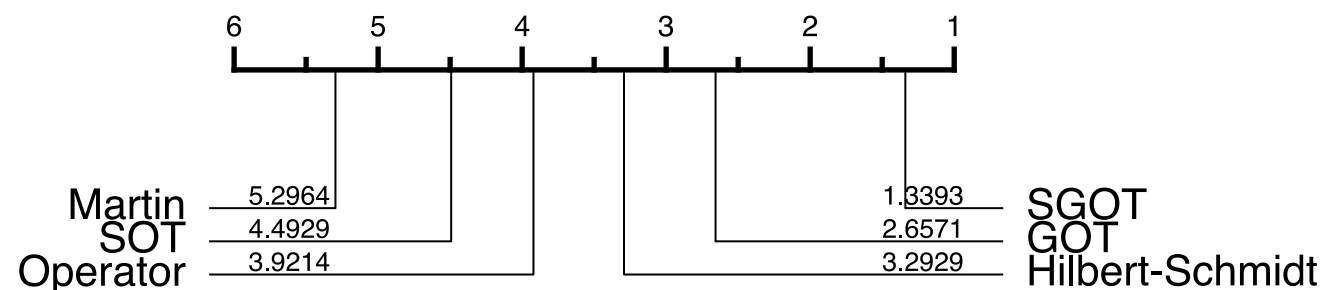
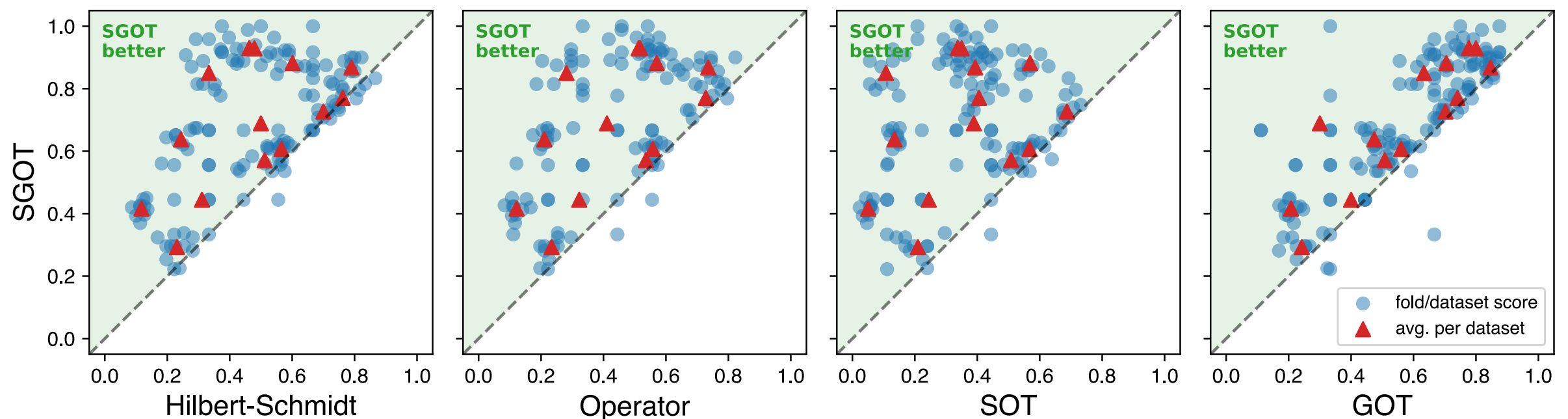
Classification

Dataset: 14 datasets from the UEA time series database.

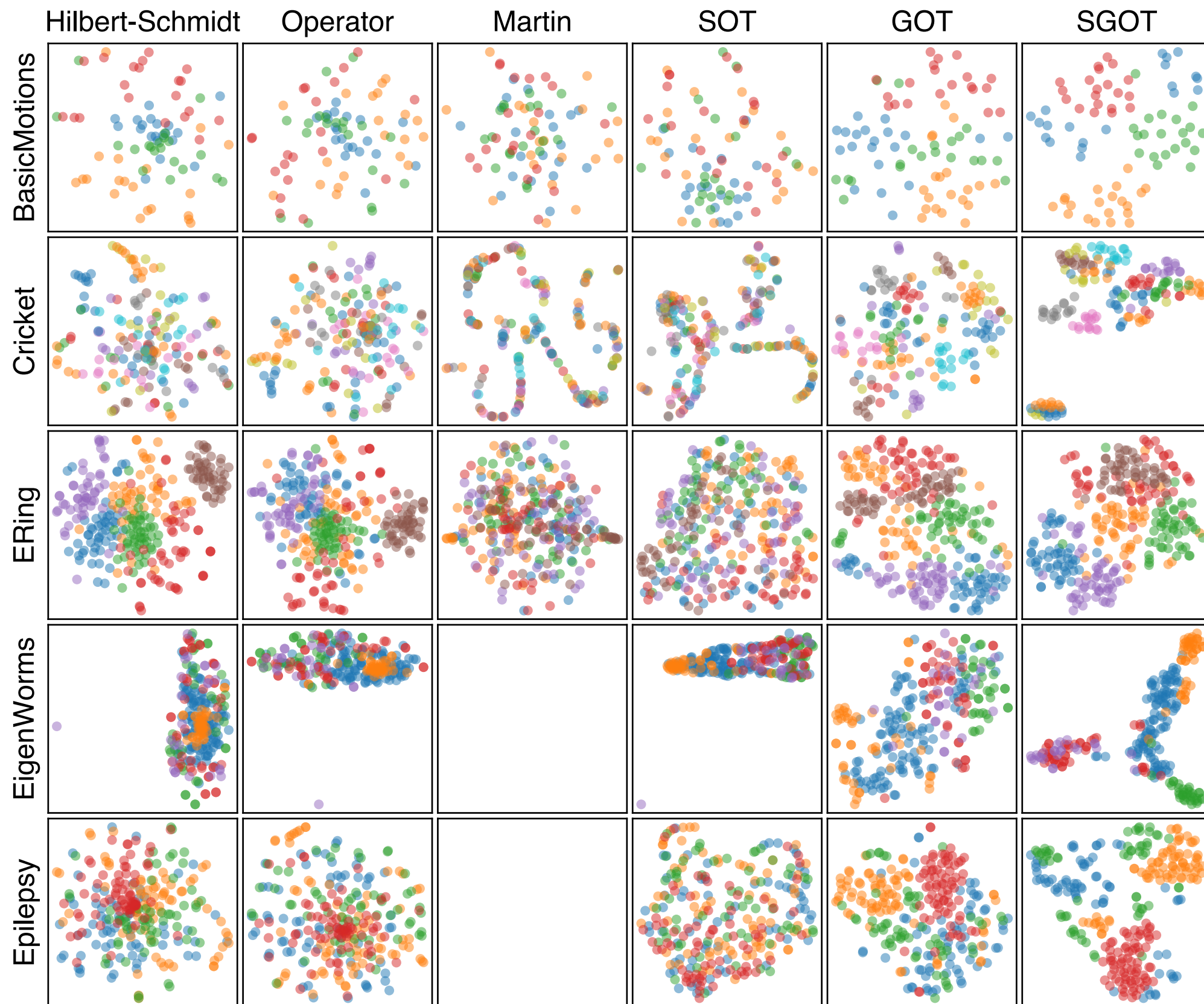
Classifier: K-nearest-neighbors with 6 different metrics.

Score function: Accuracy.

Protocol: Nested cross-validation + grid search for neighbors K and SGOT ratio η .

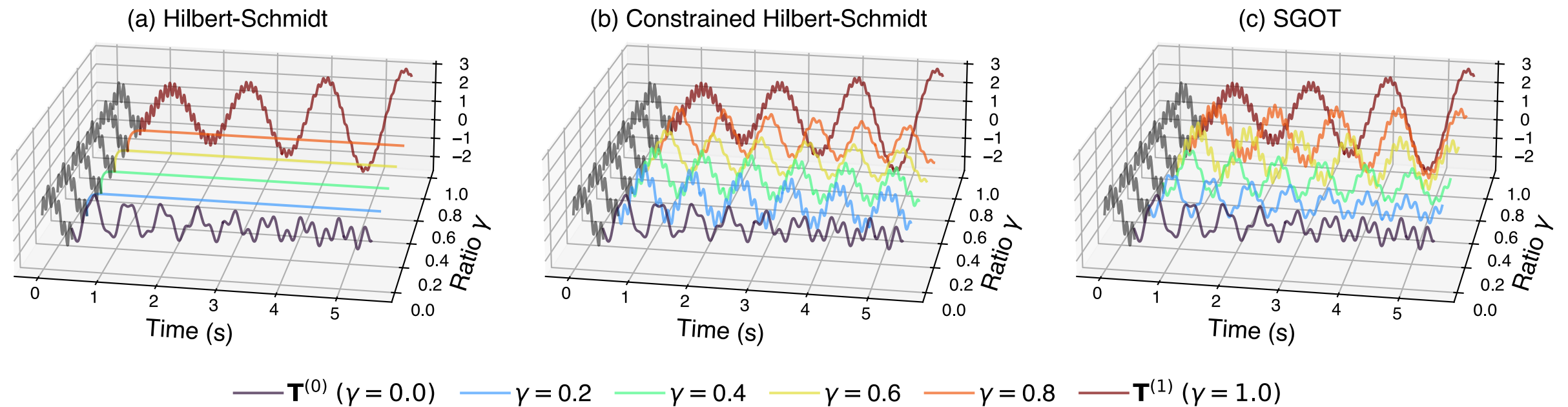


T-SNE embeddings

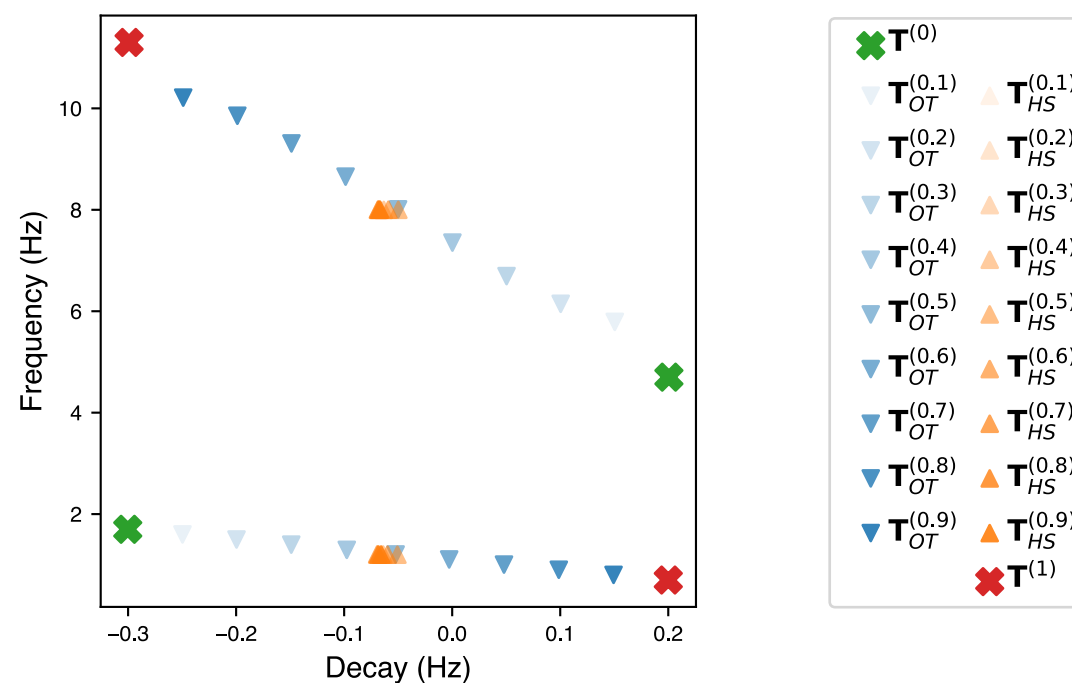


Barycenter & Interpolation

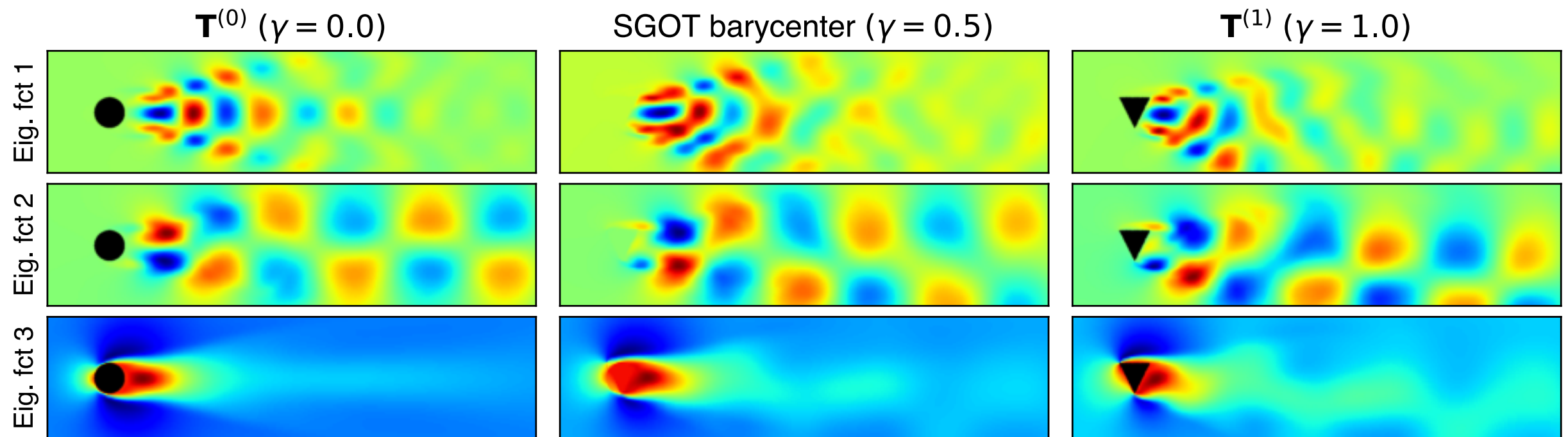
Barycentric interpolation between two linear dynamical systems.



Barycenters' eigenvalues evolution:



Barycenter & Interpolation



Experimental setup:

- Barycentric interpolation between two fluid dynamics. Fluid flow from left to right through an obstacle (circle or triangle).
- Estimate their transfer operators and compute their SGOT barycenter.
- Recovered the three first eigenfunctions for all dynamical systems (source, target, and interpolated)

Thank you